



MECH MAINT- BOILER Department
SIMHADRI

Note No :1



Ref:SMTTP/MECH MAINT-/2023-24/LMI /989106

Date:

12/03/2024(12:54:50)

Subject:Approval of New LMI “BOILER COOLING CUM PRESERVATION
PROCEDURE”

LMI No: **COS-ISO-00-OGN/OPS/SYST/059**

This is a new LMI which was prepared as per the COS-ISO-00-OGN/OPS/SYST/059 guidelines.

Put up for approval please

Submitted by:

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On:12/03/2024(12:54:50) **Ref:**SMTTP/MECH MAINT-/2023-24/LMI /989106

Note No:2

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Note No:3

A new LMI "LMI/OGN/OPS/SYST/059" has been prepared w.r.t ref OGN document "COS-ISO-00-OGN/OPS/SYST/059".

The document has been reviewed and found that all the guidelines pertaining to sub-critical boilers have been incorporated.

CA may please accord approval for this new LMI with ref no. LMI/OGN/OPS/SYST/059.

Submitted by:

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Note No:4

This is a new LMI for "Boiler Cooling cum Preservation" prepared w.r.t. OGN doc "COS-ISO-00-OGN/OPS/SYST/059".

The LMI attached as Annexure "Boiler Cooling cum Preservation Procedure " may please be approved.

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Note No:7

Review and comment, if any.

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Note No:8

New LMI is reviewed w.r.t OGN.

As per LMI, electrical maintenance of LRSB, SB, WB is covered as per usual practice.

Put up for approval please

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Note No:9

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Note No:10

Pl review the LMI and forward for approval

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Note No:11

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Note No:12

Operation department is following the above said procedure , put up for approval please

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Note No:14

Approved & Submitted by:

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On:06/04/2024(19:28:47) **Ref:**SMTTPP/MECH MAINT-/2023-24/LMI /989106

NTPC

SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

LMI/OGN/OPS/SYST/059

ISSUE NO.0; REVISION NO. 0; DATE: FEB'2024

**BOILER COOLING CUM
PRESERVATION PROCEDURE**

APPROVED FOR

***Implementation*.....**

HOP, SIMHADRI

Date-----

LMI APPROVAL DOCUMENTS

<u>1.0 TITLE OF LMI</u>	BOILER COOLING CUM PRESERVATION PROCEDURE
<u>2.a) DOCUMENT CODE</u> b) REVISION/ISSUE NO.&DATE	LMI/OGN/OPS/SYST/059 <u>ISSUE-0; REV.-0; FEB'2024</u>
<u>3.ORIGINAL/REF.DOCUMENT</u>	<u>COS-ISO-00-OGN/OPS/SYST/059</u>
<u>4.PREPARED BY:</u> SENIOR MANAGER (MM-PP)	SIGNATURE-----
<u>5. CHECKED BY:</u> AGM(MM-BMD) AGM (BMD) AGM(EMD) GM(OPERATION)	SIGNATURE----- SIGNATURE_____ SIGNATURE_____ SIGNATURE_____
<u>6. REVIEWED BY</u>	SIGNATURE:----- GM (O&M)
<u>7. APPROVED BY:</u>	SIGNATURE:----- HOP (NTPC SIMHADRI)
<u>7. DATE OF ISSUE</u>	
<u>8. INDEX LIST</u>	ENCLOSED.
<u>9. DISTRIBUTION LIST</u>	ENCLOSED.

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2.0 INTRODUCTION

The Objective of this Guidelines is to ensure the safe and effective maintenance of the boiler during shutdown periods to prevent corrosion, rusting, and other forms of deterioration. This includes procedures for gradually cooling down the boiler to avoid thermal stress, draining water and removing any remaining impurities, and applying appropriate preservation measures to safeguard critical components. The goal is to maintain the integrity and longevity of the boiler system, ensuring it can be safely restarted and operated efficiently after the shutdown period.

3.0 SCOPE

The scope of this LMI is sub critical drum boilers (Stage 1&2) at NTPC Simhadri.

4.0 REFERENCE DOCUMENT

Operation information note COS-ISO-00-OGN/OPS/SYST/059 circulated by operation services-NTPC corporate center.

5.0 SUPERSEDED DOCUMENT

Not applicable

6.0 SPECIFIC INSTRUCTIONS

The objective of this LMI is establishing the boiler cooling cum preservation procedure to minimize stresses in the boiler during the shutdown process, optimize cooling time and if required to switch for boiler preservation in case of RSD

7.0 PROCEDURE:

7.1 . Do's and Don'ts			
Sl. No.	Activities	Responsibilities	Remarks
1	No force cooling is permitted by draining boiler hot water and simultaneously taking cold water in boiler.	HOD(Operation)	
2	Maximum rate of fluid temperature drop in drum type boiler 1.4°C/min.	HOD(Operation)	
3	Maximum Drum top and bottom difference in forced circulation boilers to be kept within 50°C.	HOD(Operation)	

4	Maximum rate of fluid temperature drop in Supercritical 1.70°C/min.	HOD(Operation)	
5	Maximum rate of metal temperature drop of SH/RH tube 2.70°C/min.	HOD(Operation)	
6	Maximum feed water to metal temperature difference 111°C.	HOD(Operation)	
7	WW temperature reference to be seen from water wall thermocouples/suction manifold/highest drum bottom temperature.	HOD(Operation)	
8	In case of tube leakage in Pent house or leakage near roof in furnace, all pent house manholes including roof manholes to be opened and exhaust fans to be put in service.	HOD(BMD)	
9	Note down boiler expansions in First and second pass before stoppage and immediately after boiler draining as per annexure-A. Note: Efforts should be made to take at least bottom 4 expansion readings as early as possible during shutdown.	HOD(Operation) HOD(BMD)	
10	First goose neck LRSB and subsequently wall blower to be used to clean boiler by locked up steam and reduce steam pressure. If soot blower steam supply is from CRH then ERV to be used to reduce locked up pressure slowly.	HOD(Operation) HOD(BMD) HOD(EMD)	
11	HP/LP Bypass should not be used for depressurizing the boiler.	HOD(Operation)	
12	When main steam pressure reached 10 Kg/cm ² in pendant type SH (non-drainable SH) and 5 Kg/cm ² in horizontal type SH, open all vents valves, Separator/Drum Vents at 2 KSC to be opened and Water wall to be drained at 110°C (This is as per Boiler preservation procedure Ref: OGN/OPS/CHEM/017)	HOD(Operation)	
13	For Boiler PTW water wall draining should be as per BMD requirement. The requirement shall be given in writing by BMD and the Senior Authorised person will prepare the Safety document after considering all safety aspects. The draining status shall be mentioned in the safety document issued by Senior Authorized Person.	HOD(Operation) HOD(BMD)	
14	BMD may start boiler washing when boiler tube surface temperature comes < 100°C.	HOD(BMD)	

7.2. . Cooling procedure for boiler shutdown: In External BTL or Non-emergent shutdown (RSD)

Sl.No.	Activities	Responsibilities	Remarks
1	In External BTL or non-emergent shut down (RSD) operate first goose neck LRSB then other LRSBs and Wall blowers to clean boiler surfaces and remove ash accumulations. Soot blowers to be operated one by one before and after Shut Down (if steam supply is from CRH then ERVs may be used)	HOD(Operation) HOD(BMD) HOD(EMD)	
2	Gradually reduce load and allow pressure drop & steam temperature drop as per Turbine manufacturer recommendations.	HOD(Operation)	
3	Thereafter, stop one FD and one ID fan.	HOD(Operation)	
4	Do not open any manhole and peephole	HOD(Operation) HOD(BMD)	
5	Keep running one stream at minimum air flow (25 to 30%) till Flue gas temperature at air preheater inlet comes to 205°C. (or as per OEM recommendations whichever is lower: This is to protect damage to air preheater seals)	HOD(Operation)	
6	After stopping FD fan, Keep one ID fan in service to maintain negative 5mm draft in furnace, furnace draft may be regulated to ensure boiler cooling rate as mentioned in clause 4.1 point-2,3,4 & 5.	HOD(Operation) HOD(BMD)	
7	Open boiler manhole doors and peephole doors. Written communication in this regard from boiler maintenance shall be recorded.	HOD(Operation) HOD(BMD)	
8	Seal trough draining may be done monitoring boiler cooling rate as mentioned in clause 7.1 point-2,3,4 & 5.	HOD(Operation)	
9	Keep BRP/BCP in service till boiler draining starts.	HOD(Operation)	
10	Open Superheater vents at 10KSC. Maintain drum/separator level by feed water supply.	HOD(Operation)	
11	Open drum/separator vents at 5.0 KSC.	HOD(Operation)	
12	Boiler to be drained as per BMD requirement or as per boiler dry air preservation procedure (110°C) OGN/OPS/CHEM/017. In case of wet preservation, follow COS/OGN/Chem/017	HOD(Operation)	
13	Keep scanner air fan running till boiler draining/Cooling.	HOD(Operation)	

14	Boiler washing PTW as per BMD requirement with or without draining boiler to remove any fouling before entering the boiler.	HOD(Operation) HOD(BMD)	
15	In case of Unit RSD Follow boiler preservation guidelines as per Wet/ Dry preservation scheme for WW, SH & RH and follow OGN/OPS/CHEM/017.	HOD(Operation) HOD(BMD)	
16	In case APH washing (when actual DP is more than 1.5 times of design), guidelines as per COS-ISO-00- OGN/OPS/MECH/055 Rev. No.: 0 to be followed.	HOD(Operation) HOD(BMD)	
17	For drum type units, sufficient blow down of boiler is to be done before shutdown of unit.	HOD(Operation)	

7.3. Cooling procedure for boiler emergent shutdown:

Sl. No.	Activities	Responsibilities	Remarks
1	Boiler shut down immediately after identifying BTL as per ASLD, if ASLD not available immediately after its confirmation from local and inform BMD (This is to minimise secondary damages.)	HOD(Operation) HOD(BMD)	
2	Further steps as per above clause 7.2 point no 3. onwards.	HOD(Operation) HOD(BMD)	

7.4. Boiler Checking by filling:			
Sl. No.	Activities	Responsibilities	Remarks
1	To identify any seepages in the water circuit BMD to record communication in control room for boiler cold water filling (if required) thru boiler fill pump Maintaining recommended chemistry parameters for boiler filling (for limit, refer to Sec. 'Hydro test of boiler' of OGN/OPS/Chem/017 Clause-11.6)' (in case boiler fill pump is not available, filling may be done thru boiler feed pump, dearator pegging to be done after consultation with BMD). After completing water circuit inspection, this water is to be drained	HOD(Operation) HOD(BMD)	
2	If any seepages are observed, then after finishing work or if No seepages are observed then BMD to again record clearance for boiler hot & deaerated water filling.	HOD(BMD)	
3	Boiler fill pump system and boiler filling line to be kept in healthy condition.	HOD(Operation) HOD(BMD)	

Protocol of Clause-7 of this TC document should be signed by Operation and BMD in every BTL.

Annexure-A Expansion Indicator Records

Load	Expansion Indicator at bottommost location of 1st pass (All four corners)	Expansion Indicator at bottommost location of 2nd pass (All four corners)
Full load		
Immediate after unit shutdown		
Immediate after boiler draining		

Annexure-B Estimated MTTR

Estimated time for BTL repair - Subcritical two pass boiler (500 MW)

Location	Boiler Draining	Water washing	Scaffolding erection/ Sky climber/ inflatable jack	inspection	Attachment cutting	tube cutting, edge preparation, joint fit up, welding and stress relieving (Repair time)	Radiography/ PAUT	Restoration of support/ clamp/ scallop/ attachment/ fin etc.	Scaffolding removal/ Sky climber removal/ inflatable jack removal	Boiler filling and checking	total time upto boiler light up clearance	Ligtup to synchroni sation	Bar to Bar
Economiser making dummy or in position repair	10	3	2	1	1	4 hours for every 6 joints	2 hours for every 6 joints	2	1	2	28	3.5	31.5
Economiser making gap with inflatable jack	10	3	3	1	1	4 hours for every 6 joints	2 hours for every 6 joints	2	1	2	29	3.5	32.5
Economiser repair by lifting and lowering the coil	10	3	8	1	1	4 hours for every 4 joints	2 hours for every 4 joints	3	2	2	36	3.5	39.5
LTSH repair in position	10	3	2	1	1	4 hours for every 4 joints	2 hours for every 4 joints	2	1	2	28	3.5	31.5
LTSH repair requiring gap with inflatable jack	10	3	3	1	1	4 hours for every 6 joints	2 hours for every 6 joints	2	1	2	29	3.5	32.5
Steam cooled wall (2nd pass)	10	3	2	1	1	4 hours for every 6 joints	2 hours for every 6 joints	2	1	2	28	3.5	31.5
Superheater/Reheater	10	3	4	2	1	4 hours for every 6 joints	2 hours for every 6 joints	2	2	2	32	3.5	35.5
Superheater/Reheater T91	10	3	4	2	1	17 hours for every 6 joints	2 hours for every 6 joints	2	2	2	45	3.5	48.5
Waterwall	10	3	4	1	1	4 hours for every 6 joints	2 hours for every 6 joints	2	2	2	31	3.5	34.5
Divisional	10	3	10	2	1	4 hours for every 6 joints	2 hours for every 6 joints	1	2	2	37	3.5	40.5
Pent house	36	0	1	2	1	4 hours for every 6 joints	2 hours for every 6 joints	1	1	2	50	3.5	53.5
Pent house with PWHT	36	0	1	2	1	17 hours for every 6 joints	2 hours for every 6 joints	1	1	2	63	3.5	66.5

Estimated time for BTL repair - Subcritical two pass boiler (210 MW & below)

Location	Boiler Draining	Water washing	Scaffolding erection/ Sky climber/ inflatable jack	inspection	Attachment cutting	tube cutting, edge preparation, joint fit up, welding and stress relieving (Repair time)	Radiography/ PAUT	Restoration of support/ clamp/ scallop/ attachment/ fin etc.	Scaffolding removal/ Sky climber removal/ inflatable jack removal	Boiler filling and checking	total time upto boiler light up clearance	Lightup to synchronisation	Bar to Bar
Economiser making dummy or in position repair	9	3	1.5	1	1	4 hours for every 6 joints	2 hours for every 6 joints	2	1	1	25.5	3	28.5
Economiser making gap with inflatable jack	9	3	2.5	1	1	4 hours for every 6 joints	2 hours for every 6 joints	2	1	1	26.5	3	29.5
Economiser repair by lifting and lowering the coil	9	3	4	1	1	4 hours for every 4 joints	2 hours for every 4 joints	3	2	1	30.0	3	33.0
LTSH repair in position	9	3	2	1	1	4 hours for every 4 joints	2 hours for every 4 joints	2	1	1	26.0	3	29.0
LTSH repair requiring gap with inflatable jack	9	3	3	1	1	4 hours for every 6 joints	2 hours for every 6 joints	2	1	1	27.0	3	30.0
Steam cooled wall (2nd pass)	9	3	2	1	1	4 hours for every 6 joints	2 hours for every 6 joints	2	2	1	27.0	3	30.0
Superheater/Reheater	9	3	3	2	1	4 hours for every 6 joints	2 hours for every 6 joints	2	2	1	29.0	3	32.0
Superheater/Reheater T91	9	3	3	2	1	12 hours for every 6 joints	2 hours for every 6 joints	2	2	1	37.0	3	40.0
Waterwall	9	3	3	1	1	4 hours for every 6 joints	2 hours for every 6 joints	3	2	1	29.0	3	32.0
platen	9	3	6	2	1	4 hours for every 6 joints	2 hours for every 6 joints	1	2	1	31.0	3	34.0
Pent house	24	0	1	2	1	4 hours for every 6 joints	2 hours for every 6 joints	1	2	1	38	3	41.0
Pent house with PWHT	24	0	1	2	1	15 hours for every 6 joints	2 hours for every 6 joints	1	2	1	49	3	52.0